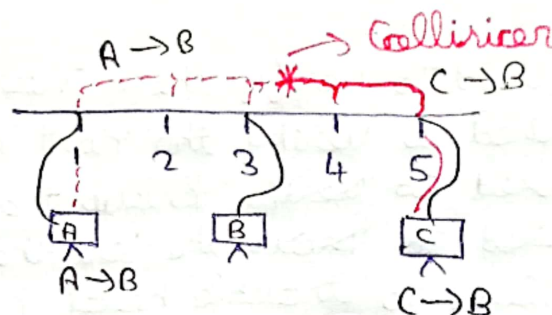


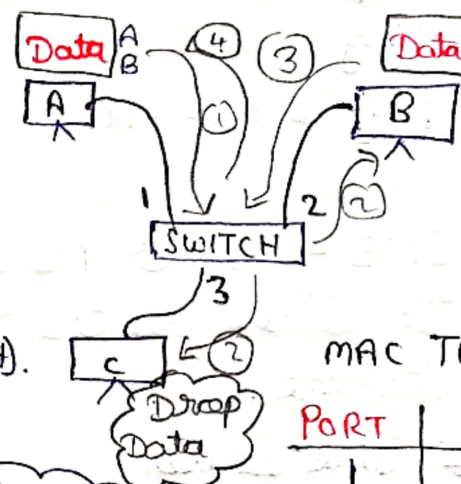
### 1, Explain behaviour of hub device

- \* Hub is a broadcast device.
- \* It produces heavy traffic.
- \* Collision occurrence is high.
- \* One-way Communication device.
- \* Layer 1 Device.
- Eg: Walkie-Talkie



### 2, Explain behaviour of switch device

- \* Switch is a Layer 2 device.
- \* It has a MAC table which stores the Sender Port and its MAC address.
- \* If packet arrives, it checks the MAC table, if found the forwards it (unicast).
- \* Else creates a broadcast and stores the port and MAC details in table.
- \* This technique is known as "Address Learning".
- MAC table life time is 5 min.



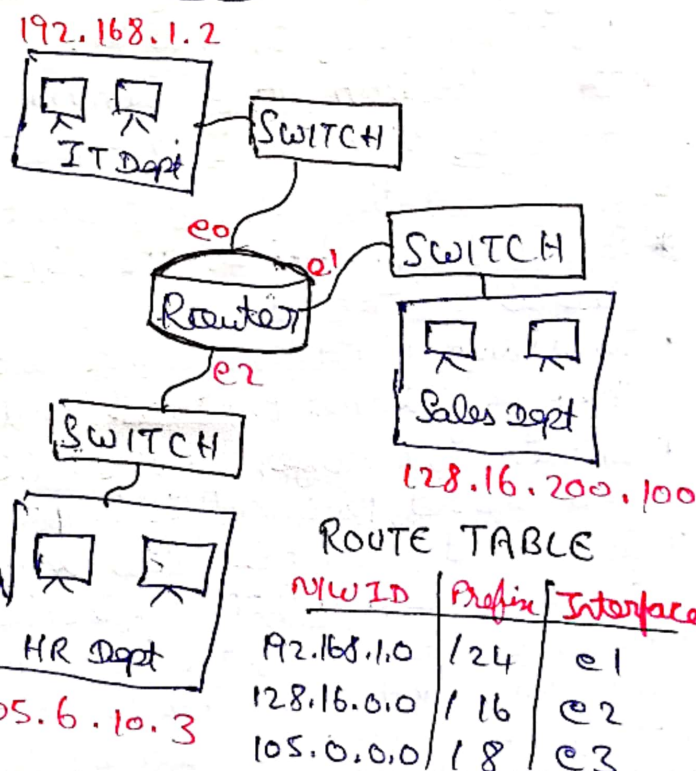
MAC TABLE

| PORT | MAC |
|------|-----|
| 1    | A   |
| 2    | B   |



### 3, Explain behaviour of router device

- \* It is a Layer 3 device.
- \* It is used to breakdown the Single broadcast domains.
- \* It maintains Routing Table and this is called Routing.
- \* Routing table n/w ID, Prefix and Interface.
- \* If a data packet is arrived it checks the routing table and forwards only to the respective IP.
- \* It is a Unicast device.



ROUTE TABLE

| N/W ID      | Prefix | Interface |
|-------------|--------|-----------|
| 192.168.1.0 | /24    | e1        |
| 128.16.0.0  | /16    | e2        |
| 105.0.0.0   | /8     | e3        |